



Prediction of monkeypox infection from clinical symptoms with adaptive artificial bee colony-based artificial neural network

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Abstract

In 2022, the World Health Organization declared an outbreak of monkeypox, a viral zoonotic disease. With time, the number of infections with this disease began to increase in most countries. A human can contract monkeypox by direct contact with an infected human, or even by contact with animals. In this paper, a diagnostic model for early detection of monkeypox infection based on artificial intelligence methods is proposed. The proposed method is based on training the artificial neural network (ANN) with the adaptive artificial bee colony algorithm for the classification problem. In the study, the ABC algorithm was preferred instead of classical training algorithms for ANN because of its effectiveness in numerical optimization problem solutions. The ABC algorithm consists of food and limit parameters and three procedures: employed, onlooker and scout bee. In the algorithm standard, artificial onlooker bees are produced as much as the number of artificially employed bees and an equal number of limit values are assigned for all food sources. In the advanced adaptive design, different numbers of artificial onlooker bees are used in each cycle, and the limit numbers are updated. For effective exploitation, onlooker bees tend toward more successful solutions than the average fitness value of the solutions, and limit numbers are updated according to the fitness values of the solutions for efficient exploration. The performance of the proposed method was investigated on CEC 2019 test suites as examples of numerical optimization problems. Then, the system was trained and tested on a dataset representing the clinical symptoms of monkeypox infection. The dataset consists of 240 suspected cases, 120 of which are infected and 120 typical cases. The proposed model's results were compared with those of ten other machine learning models trained on the same dataset. The deep learning model achieved the best result with an accuracy of 75%. It was followed by the random forest model with an accuracy of 71.1%, while the proposed model came third with an accuracy of 71%.

Keywords Monkeypox · Monkeypox clinical symptoms · Machine learning · Artificial neural network · Artificial bee colony algorithm · Adaptive artificial bee colony algorithm

1 Introduction

In 2022, the monkeypox virus (MPXV) spread, causing panic among people, and causing concern among scientists due to its rapid spread [1]. Approximately 1–11% of cases lead to death [2]. The World Health Organization (WHO) announced that the number of people infected with this virus had increased significantly, and it was confirmed that more than 318,000 patients were infected in August 2022

[3]. This virus belongs to the genus of corticoviruses and is similar to zoonotic smallpox [4]. It is caused by the orthopoxvirus and is a genera of the poxviridae family that is dangerous to humans [5]. Figure 1 shows infection with monkeypox, which begins to appear after 3 days at most of the infection with the fever. At first, the symptoms are on the face, and then, they spread to the rest of the body. A person is considered contagious before the rash appears by five days. And it remains until a new layer of skin forms underneath. It takes between two to four weeks.

The disease was detected for the first time in Africa, specifically in the Republic of the Congo [4]. And then it spread among the countries of the world. As of June 2022, more than 1,256 cases of monkeypox have been reported in

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Fig. 1 Symptoms of infection in monkeypox disease that appear on the skin [41]

several regions of Spain. Statistics indicate that the majority of those infected are males. At the same time, the average age was about 36 years [6]. The method of infection with monkeypox is direct contact with an infected person, animal or other material. It is also transmitted through the mucus of the nose, mouth or eyes [7]. And having sex is one of the ways the virus is transmitted. [5]

The clinical picture of monkeypox and smallpox is very similar, and the symptoms that appear upon infection differ from one case to another. However, skin rash is the most likely sign of infection, along with anogenital lesions, lethargy and muscle pain [7]. Monkeypox symptoms last up to four weeks. Children are also the most vulnerable [8]. Patients with the disease may suffer several side effects, including bronchiolitis, hypothermia, bacterial infections and respiratory failure [9]. Diagnosing the condition based on a range of clinical features is difficult. An accurate diagnosis of monkeypox requires a molecular test in a specialized laboratory to distinguish it from other diseases. The appearance of the first positive PCR result may take 5 days from the onset of symptoms [7].

Therefore, early detection of the disease is crucial to control transmission. It is also necessary to continue describing the symptoms of the disease and its transmission mechanisms to reduce its risk among populations [10].

With the spread of artificial intelligence applications, researchers have resorted to using it in diagnosing disease conditions in medical and biomedical applications [11]. They used it in multiple ways, depending on the dataset collected from the lesions' images or the infected's clinical symptoms.

In the field of automatic virus identification in transmission electron microscopy (TEM) images, [11] relied on image datasets to characterize the monkeypox virus. It consists of 1245 micrographs of 22 viruses taken by TEM. However, this study was limited to 14 types of viruses, such as Astrovirus & Adenovirus & CCHF & Ebola, etc. The study used convolutional neural network (CNN) deep learning (DL) models to build its model. The accuracy rate of the proposed method was 93.1%.

[2] Use different approaches in data acquisition. The data is a set of images of skin lesions. Collected by manual searches and contact with infected persons. The study focused on separating monkeypox from similar cases of different types of smallpox. The approach taken was VGG16 deep transfer learning. It consists of three layers of convolutional filters to extract the features from the images and then the neural network. It was a perfect idea that he used transfer learning. The accuracy rate of the results obtained was 86%.

[12] Research was divided into three separate studies. All of them were conducted on the proposed approach. Transfer learning approaches (GRA-TLA) work on multi-class classification using generalization and regularization. The training dataset is the images of skin lesions. It was intended to support decision-making assistance to the hospital. Computational results showed that the proposed approach could distinguish between infected and non-infected monkeypox individuals with an accuracy of 77–88% in the first and second studies. At the same time, the residual network (ResNet) had the best performance for multiclass classification in the third study, with an accuracy rate ranging from 84 to 99%.

[8] Also relied on training data consisting of images to establish an early detection mechanism for monkeypox that would help identify infected people. The approach taken in this paper was to compare several models of ResNet50, EfficientNetB3 and EfficientNetB7 algorithms. In the end, it was concluded that the results of the EfficientNetB3 algorithm were the best.

Saleh and Rabie took a different approach than previous research, focusing on numerical data for clinical symptoms of the disease from 500 suspected cases from Spain and Nigeria [13]. The dataset was not limited to monkeypox but included several other diseases such as acne, alopecia, normal, psoriasis and smallpox. The human monkeypox diagnosis (HMD) strategy was applied in the study. In fact, the proposed method consisted of two phases. First: extracting the appropriate features, using the improved binary chimp optimization (IBCO) algorithm, which is a

hybrid selection algorithm. The second stage: composed of three machine learning algorithms weighted naïve Bayes (WNB), weighted k-nearest neighbor's (KNN) and deep learning. In fact, a final election is made for the output of these three algorithms.

Computer-aided healthcare has become an essential field for artificial intelligence researchers as it provides the advantage of early diagnosis for many diseases [14]. Medical data (especially images) required for disease diagnosis can be processed with artificial intelligence algorithms, and disease diagnosis can be made. When the literature studies are examined, it is noticed that the proposed methods for the early detection of monkeypox also focus on image processing techniques. However, the fact that medical data is generally obtained in chronological order and cannot be collected simultaneously makes it challenging to apply image processing techniques [15]. In addition, although image processing techniques produce successful solutions to problems with significant differences, they may fail in classification mainly due to the close similarity between similar classes [16]. In this context, taking a different approach from image processing, Saleh and Rabie used a numerical dataset of disease symptoms. However, in their study, they focused only on the detection of monkeypox and similar diseases. This paper presents a predictive model that will enable early detection of monkeypox disease. Unlike other studies, analysis data on disease symptoms were discussed in the study instead of images taken from diseased individuals. The artificial neural networks (ANN) were used to diagnose and predict disease. However, the artificial bee colony (ABC) algorithm was used instead of classical learning algorithms for network training due to its effectiveness in numerical optimization problems. In addition, the adaptive model (aABC for short) was preferred for faster convergence, not the standard version of the ABC algorithm. The study offers a novel approach to the literature with an ANN model trained with an adaptive metaheuristic approach for the early detection of monkeypox disease. The method's performance has been tested on clinical data from the bmj center [9] and compared with KNN, SVC, deep learning and random forest algorithms. The findings demonstrate that the proposed method can be used for the early diagnosis of monkeypox disease.

The structure of this paper is organized as follows: Sect. 2 describes the approach taken in this paper. Section 3 presents experimental studies. Findings and discussions are presented in Sect. 4. Section 5 reports on the conclusion.

2 Methodology

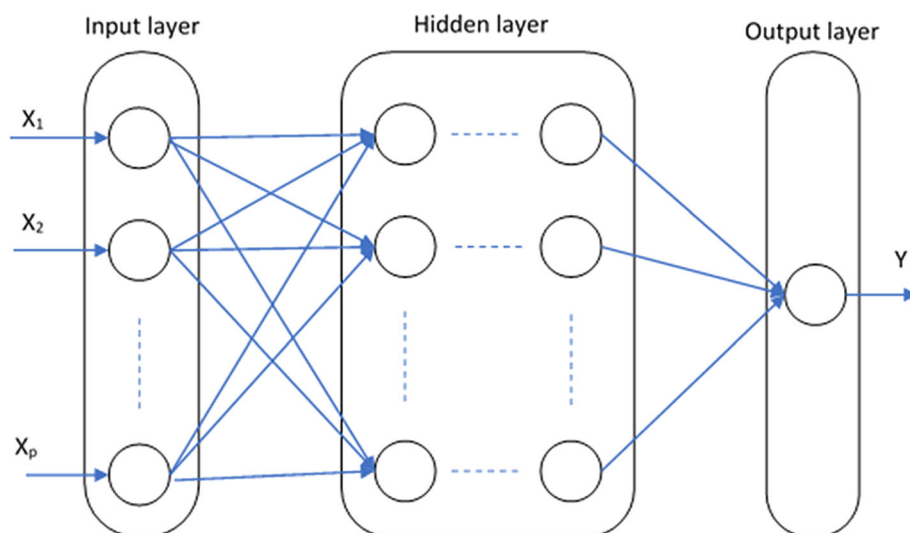
This section describes in detail the proposed method for the early detection of monkeypox. First, the classical ANN model, the model's training and the Levenberg–Marquardt learning algorithm, which is widely used, are mentioned. Since the proposed method involves training the ANN model with a metaheuristic approach, the ABC algorithm, which produces successful solutions for numerical optimization problems, is described. Next, the aABC approach, which updates the ABC parameters during the search process, is presented as an adaptive version of the algorithm. Finally, the section explains how the ANN is trained with the aABC algorithm.

2.1 Artificial neural network (ANN)

ANN model used to diagnose monkeypox patients will be described in detail. Artificial neural networks (ANN) are one of the most widely applied classification techniques for solving prediction problems [17]. The principle of ANN is based on the analysis of the biological nervous systems of an organism. ANN consists of a set of nodes and a number of interrelated processing components. ANN uses learning algorithms to simulate knowledge and store this knowledge in weighted connections, which reflect the activity of the human brain [18]. ANN architecture consists of three main components: input layers, hidden layers and output layers (Fig. 2). ANN model must be trained first in cases with known classes [19].

The input layer in the neural network receives input signals. Each neuron in the input layer corresponds to a specific input parameter. This layer passes the neural network's input to the hidden layers. The number of hidden layers varies from one model to another. The number of hidden layers is determined empirically. Neurons receive the signals coming from the first layers, and then process them by some nonlinear function of their total inputs. Each neuron refers to a real number that indicates its contribution to the output. The neuron has a weight that is adjusted during the network training process. This weight affects the neuron's contribution by multiplying it by the cell's input. The activation function of the neuron differs from one layer to another, depending on the nature of the problem to be solved. The number of output layer neurons is determined by the nature of the problem, whether it is a binary classification or more. The output of the output layer is the output of the neural network.

Fig. 2 Structure artificial neural network (ANN)



2.2 Training of ANN

Training a neural network involves adjusting the weights of neurons to improve the prediction or classification process in a task. This process is done using a training algorithm and a set of training data [20]. The process of training a neural network consists of several stages, which are as follows:

- First, Initialization: the weights and their biases are randomly initialized.
- The next stage, the forward propagation stage: the input data is passed into the network, and the activation values for each neuron in the network are calculated using weights and biases.
- Loss calculation: The output of the input data is predicted, compared with the real values, and the difference between them is calculated as a loss.
- Back propagation: the loss propagates back through the network to calculate the loss gradient at each layer and at each neuron.
- Finally, the weights and biases are updated using gradient, and the process is repeated until the loss is minimized.

There are several algorithms that can be used for training ANNs, including Adam optimization, stochastic gradient descent (SGD) and backpropagation [21]. The choice of algorithm depends on the complexity of the network and the type of task it is being trained for. It is also important to note that the quality and quantity of the training data can have a significant impact on the performance of the network [22].

2.3 Levenberg–Marquardt algorithm (L–M)

The Levenberg–Marquardt algorithm is an optimization algorithm commonly used to train artificial neural networks. It is a modification of the Gauss–Newton algorithm and is designed to efficiently solve nonlinear least-squares problems. The basic idea behind the Levenberg–Marquardt algorithm is to use a combination of gradient descent and Gauss–Newton methods to update the network weights and biases during training [23]. The algorithm calculates the Hessian matrix, which describes the curvature of the error surface, and combines it with a damping factor to prevent the algorithm from taking excessively large steps during optimization. The Levenberg–Marquardt method adaptively switches between the Gauss–Newton update and the gradient descent update when updating parameters. In Marquardt’s update relationship as Eq. 1, the damping parameter λ is scaled by the diagonal of the Hessian $J^T W J$ for each parameter.

$$[J^T W J + \lambda \text{diag}(J^T W J)] h_{lm} = J^T W (y - \hat{y}) \quad (1)$$

The algorithm adjusts the damping factor during training to balance the trade-off between convergence speed and stability. It starts with a high damping factor to provide stability during early stages of training and gradually decreases it to allow faster convergence. Compared to other optimization algorithms, the Levenberg–Marquardt algorithm typically converges faster and more accurately, especially when dealing with highly nonlinear problems. However, it can be computationally expensive and may require careful tuning of the damping factor to achieve optimal results [24].

2.4 Artificial bee colony (ABC)

It falls under swarm intelligence algorithms. Inspired by nature, mimics the work of bee swarms [25].

Swarm of bees are often concentrated in the field the most flowering. The bees reach this region by applying the swarm optimization algorithm [26]. That is, they spread initially in the field so that each bee records the area with the most flowers. Then, each bee moves randomly, and if it finds a denser area, it updates its information and so on. Upon completion of the random search, each bee announces what it has found. And then the bee swarm selects the best location [27].

It is one of the methods of artificial intelligence in meta-heuristic research problem [28]. ABC is an effective tool in finding and improving solutions. In general, its work can be summed up in finding and exploiting sources of food and then looking for a new alternative. In ABC algorithm, food reflects the initial solutions through which bees will look for the perfect solution. The quality of the food fitness represents the assessment of the solution, while the limit factor indicates the amount of food available in the source. However, cycle represents the number of searches (S. [29]).

Three groups of honeybees carry out foraging activities: (a)- Scout bees: Their duty is to find random food sources (X). (b)- Employed bees: Their number equals the number of food sources. Each employed bee moves from available resources to a specific source. She also performs a local search using Eq. (2) to find a new resource next to the resource it went to. (c)- Onlooker bees: The number of onlooker bees equals the number of food sources [30]. When the employed bees return to the hive, they watch each one's waggle dance and determine the appropriate food source to go to.

$$v_{ij} = x_{ij} + \phi_{ij}(x_{ij} - x_{kj}) \quad (2)$$

In Eq. (2), x_i represents the solution i that the employed (or onlooker) bee will be interested in. This solution for employed bees, i , means solution, but onlooker bees determine the solution they will be interested in by the choice they make with the roulette wheel. j is a randomly chosen item of the solution, ϕ is a randomly chosen coefficient in the range $[-1, 1]$, and x_k is a randomly chosen solution from among the current solutions.

The basic steps of making the ABC algorithm can be summarized as follows:—Generate an initial solution group.—Sending employed bees to food sources.—Sending onlooker bees to the most appropriate source of food.—Save the optimal source.—Repeat previous steps. Figure 3 shows the working diagram of ABC algorithm.

2.5 Adaptive artificial bee colony (aABC)

aABC algorithm overcomes some of the problems faced by ABC algorithm. It depends on the nature of the problem to be solved. In fact, aABC agrees with ABC in the main bee divisions, employed bees, scout bees and onlooker bees. However, the modification that occurs in the mechanism of action of both onlooker bees and scout bees was observed (Inspired by [31]).

First Amendment: Onlooker bees select the best food source from a group of foods accessed by employed bees. However, onlooker bees can evaluate the solution based on the specific fitness of each solution according to the following Eq. 3:

$$p_i = \frac{\text{fitness}(x_i)}{\sum_{j=1}^n \text{fitness}(x_j)} \quad (3)$$

Then, the mean fitness of all solutions is calculated. Onlooker bees only search for solutions with greater than mean fitness. The goal of this process is to search further in the set of solutions with low fitness. Hence giving it more importance.

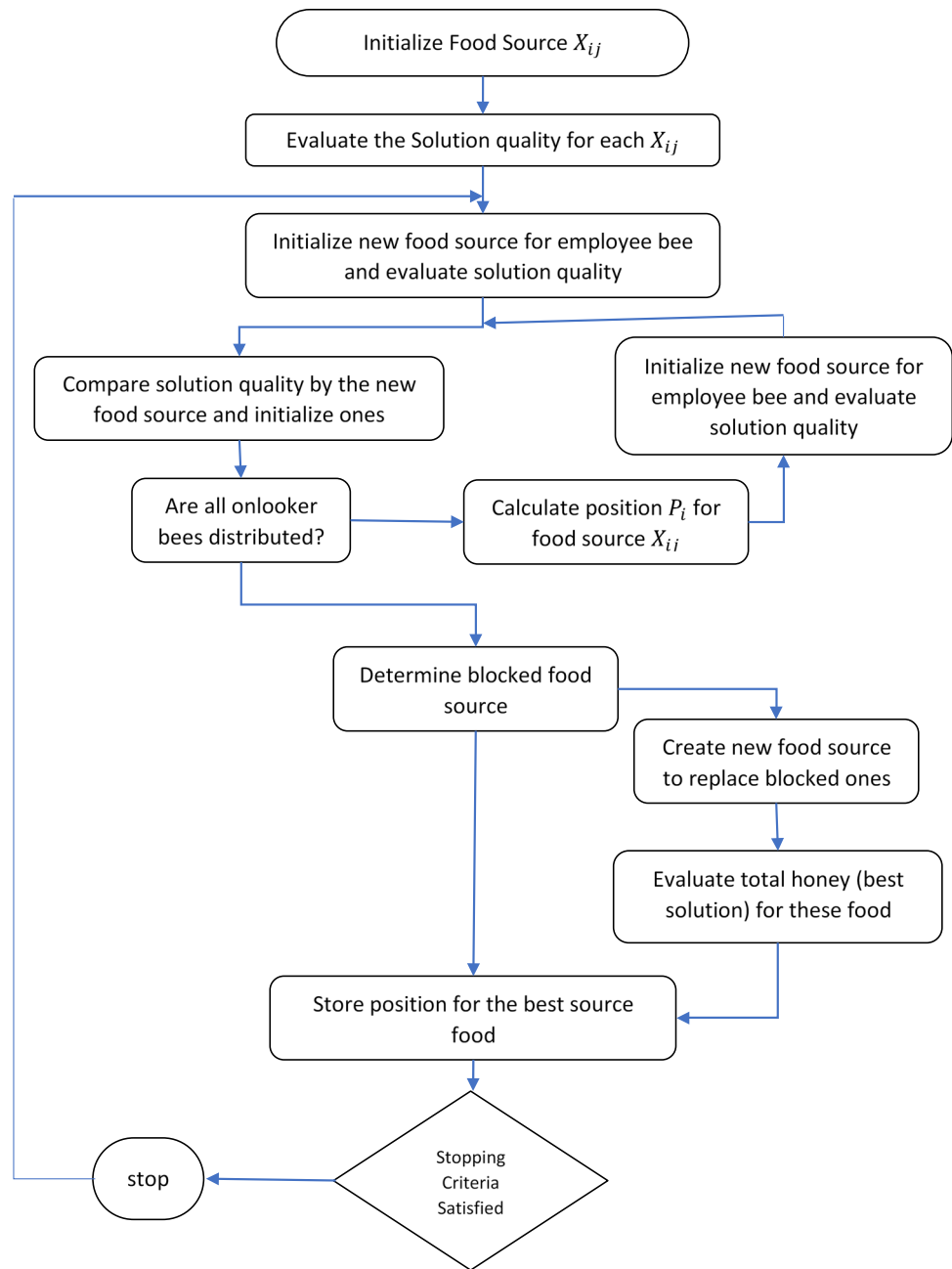
Second Amendment: Scout bees remove spent solutions from the entire population. The depletion of the solution is calculated by limit factor. In fact, each solution is assigned limit value depending on the fitness of the solution. According to the following Eq. 4:

$$\text{Limit}[i] = \frac{\text{fitness}[i] * \text{food} * D}{\sum_{i=1}^n \text{fitness}[i]} \quad (4)$$

Because the bee algorithm gives high results in search issues, these changes have affected the accuracy of the algorithm.

To solve a specific problem f , assume the computational time complexity of evaluating the function value of $O(f)$. The maximum cycle number of iterations was set to MCN and colony size to CS. The time complexity of classical ABC for this problem is $O(\text{MCN} * (\text{CS} * f + \text{CS} * (f))) = O(\text{MCN} * \text{CS} * f)$. So, the time complexity in the initialization phase is $O(\text{CS} * f)$, and the time complexity in the iterative process in the employed and onlooker bee procedures is $O(\text{MCN} * (\text{CS} * f + \text{CS} * f))$ [32]. So, standard ABC's total computational time complexity is $O(\text{CS} * f)$. aABC shares the total limit value used in standard ABC according to the fitness value of the solutions instead of dividing them equally among the solutions. So, the computational time complexity of aABC is the same with the standard ABC.

Fig. 3 Flowchart of the work of ABC algorithm [30]



2.6 Proposed model

It aims to take advantage of the ability of aABC to find the optimal solutions, in training the weights of the ANN neural network [33]. Figure 4 shows the method for training ANN weights using aABC.

The proposed model is a neural network composed of one input layer, two hidden layers and one output layer. This network is trained by aABC algorithm, which is as follows:

- Generate an initial population to search for weights. At this stage, all the neural network weights are arranged

in the form of vectors, each cell in this vector represents a specific weight of the neural network weights. This vector takes arbitrary values (the initialization of the neural network weights). However, population size is related to food factor, which determines the number of vectors that will be generated. The length of a single vector is defined by D , which is the number of weights of the neural network.

- Each vector is evaluated using the RMSE equation as in Eq. 5. Vector evaluation is calculated after all the training data has been passed and the resulting error is calculated. In fact, each vector has its own fitness.

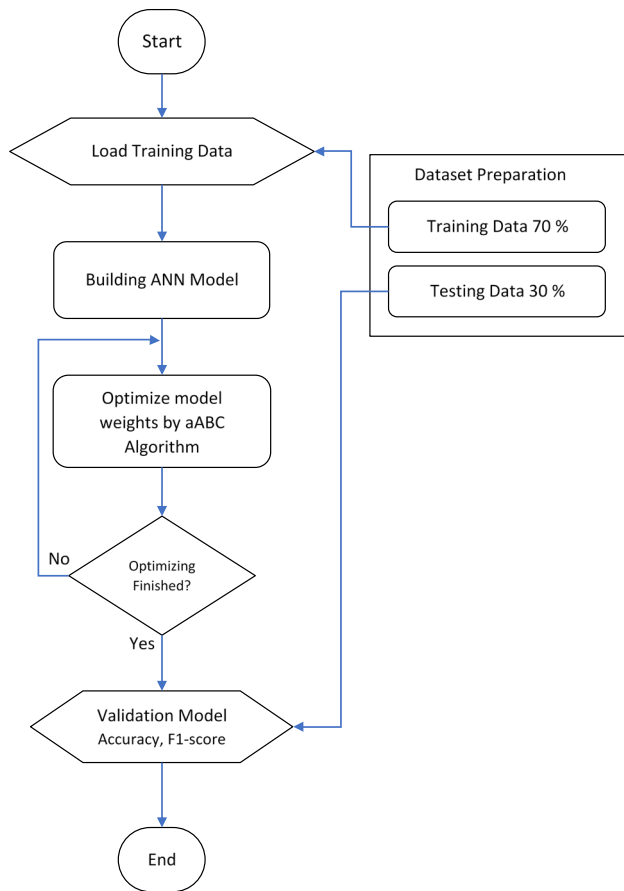


Fig. 4 Flow diagram of methodology

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (y_o - y_p)^2}{N}} \quad (5)$$

- The weights training process follows the previously mentioned bee algorithm methodology. The number of training times is subject to the Epoch factor.
- Vector with the least error value is selected and saved. In each Epoch cycle, it is considered as the ideal solution within this cycle. In fact, this cycle's best vector is compared with the previous best vector, and the vector with the lowest fitness is retained.
- At the end of the training, best vector representing the weights of the neural network is obtained among all cases.
- The final neural network is evaluated, verified and measured for accuracy. Figure 6 shows the mechanism of the proposed model.

3 Experimental study

aABC was coded in the .Net platform in C# 2022 programming language and ran on a computer with an Intel(R) Core (TM) i3 6006U 2.00 GHz processor, 12 GB RAM and Windows 10 Pro 64-bit operating system. In this section, the performance of the proposed method is tested on CEC 2019 test suites, and the method's success is compared with classical ABC and other metaheuristic methods. The section also describes the dataset used for MonkeyPox disease detection. Then, some features of the proposed model, performance measurements and assigned hyperparameter values are mentioned.

3.1 Testing on CEC 2019 test suite

The CEC 2019 test suite, called the '100 Digit Challenge', includes ten multi-modal functions designed to represent complex optimization problems. Minimization problems characterize these functions and are scalable. Additionally, it mentions that some functions (CEC01-CEC03) remain unaltered, while others (CEC04-CEC10) undergo shifts and rotations. Moreover, competitors can modify up to two parameters within the functions. The information is concise and well structured, providing a good understanding of the nature of the test suites.

The success of aABC on the CEC2019 suite has been evaluated based on the results given in [34]. Therefore, for a fair comparison, the parameter values of the ABC and aABC algorithms were determined by the parameter set in ([34]). Accordingly, the *food* is assigned as 15 (in this case, the CS is 30), and the *limit* is assigned as $\text{food} \times D$. The comparison algorithms were run independently 30 times for each pattern, and MCN in each experiment was assigned to 500. For each function's solution, search range and dimensions (D) are given in Table 4.

Other metaheuristics compared to aABC include: the flower pollination algorithm for global optimization (FPA) [35], gray wolf optimizer (GWO) [36], salp swarm algorithm (SSA) [37], cuckoo search (CS) [38], sine cosine algorithm (SCA) [39] and adaptive flower pollination algorithm (AFPA) [34].

3.2 Aabc's discussion of statistical performance

For each CEC 2019 test problem, the solution search range and dimensions (D) are given in Table 4. These algorithms were run independently 30 times for each function, and the maximum cycle number (MCN) in each experiment was assigned to 500. The following figures show the results of tests on the previously mentioned algorithms.

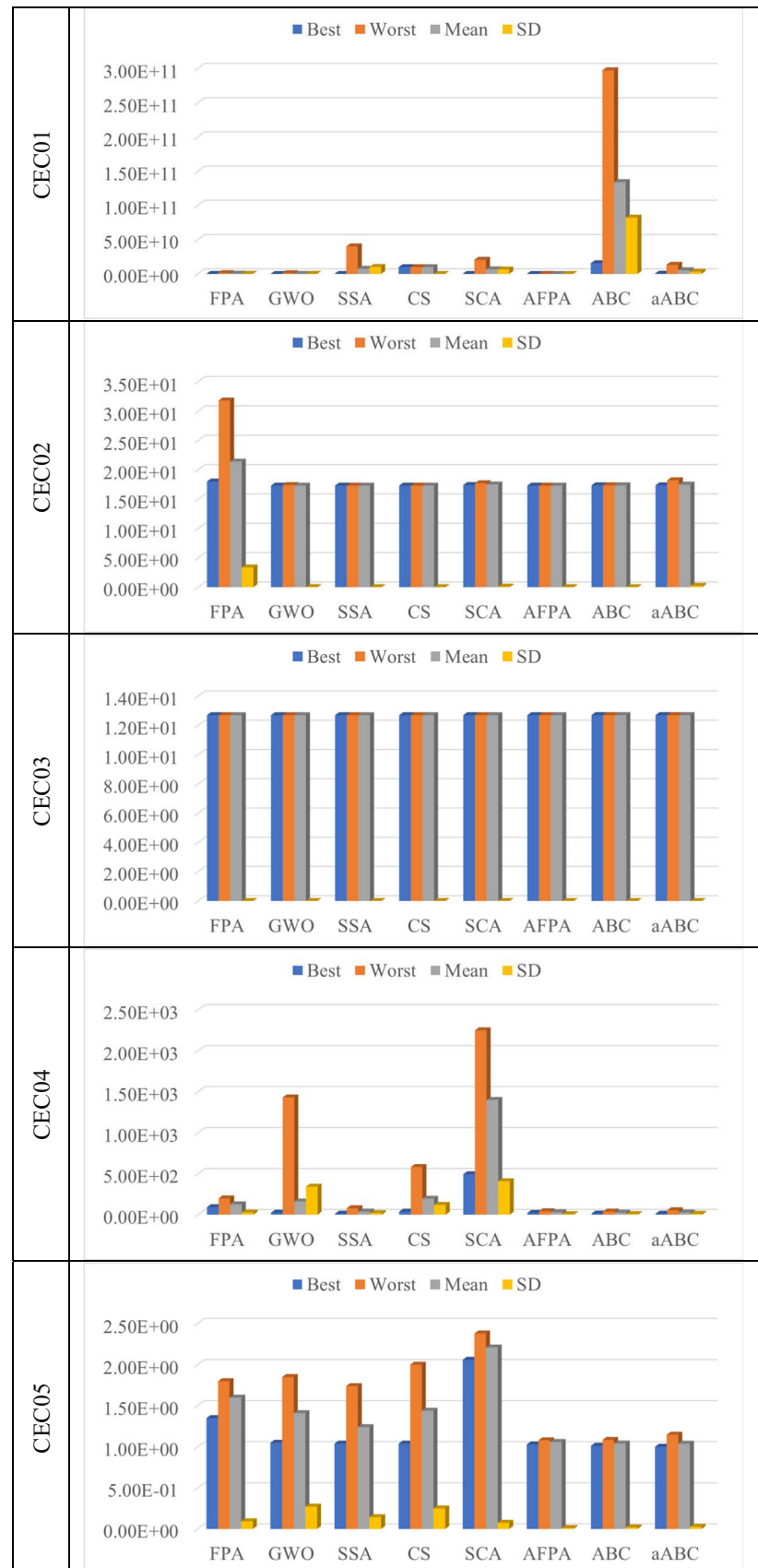
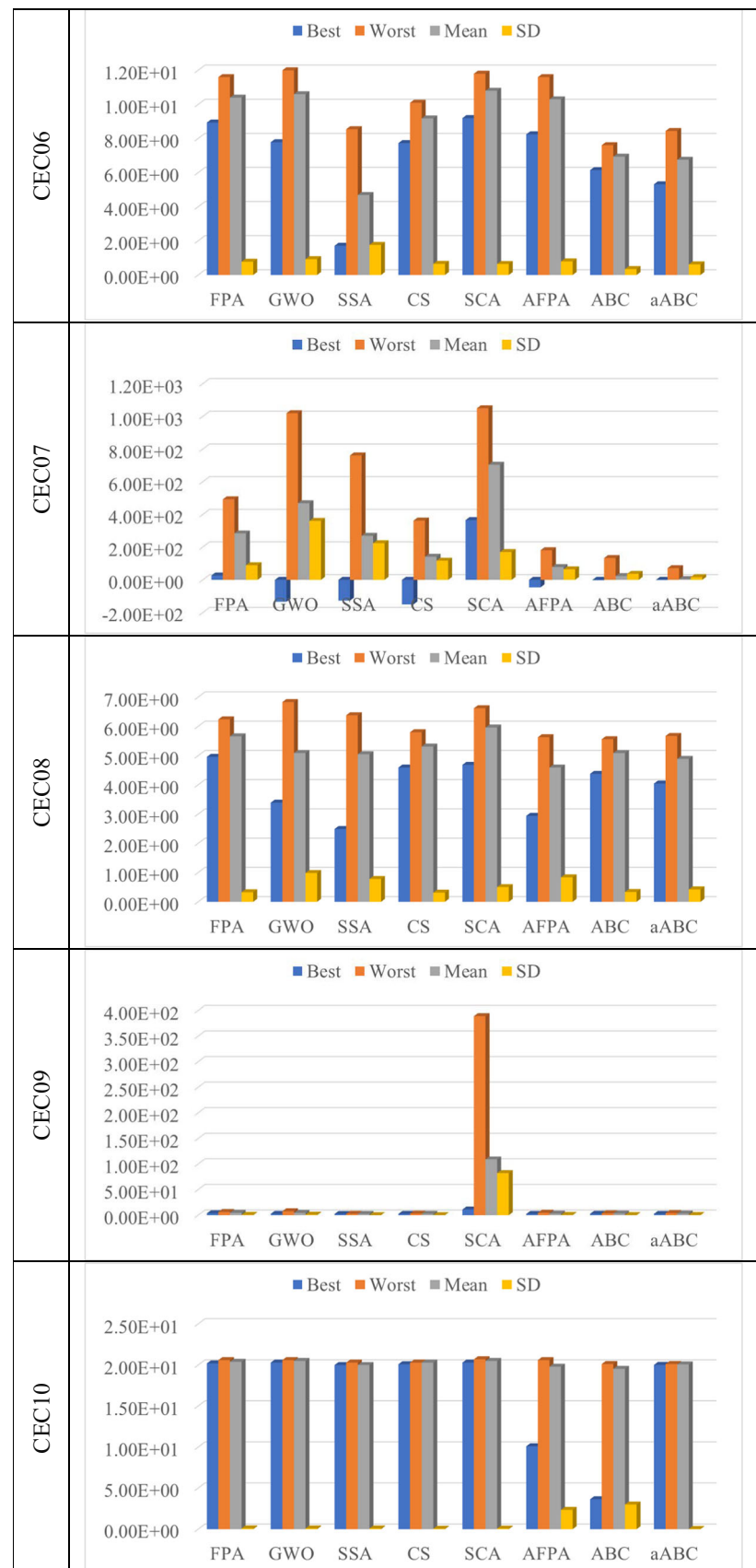
Fig. 5 Time-complexity results of algorithms

Fig. 5 continued



Patient_ID	Systemic Illness	Rectal Pain	Sore Throat	Penile Oedema	Oral Lesions	Solitary Lesion	Swollen Tonsils	HIV Infection	Sexually Transmitted Infection	MonkeyPox
P0	None	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	Negative
P1	Fever	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	Positive
P2	Fever	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	Positive
P3	None	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	Positive
P4	Swollen Lymph Nodes	TRUE	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE	Positive
P5	Swollen Lymph Nodes	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Negative
P6	Fever	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	Positive
P7	Fever	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	FALSE	Positive
P8	Muscle Aches and Pain	FALSE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	Positive
P9	Fever	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	Negative
P10	Muscle Aches and Pain	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	TRUE	Negative
P11	Swollen Lymph Nodes	TRUE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	Negative
P12	Fever	TRUE	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	Positive
P13	Swollen Lymph Nodes	TRUE	TRUE	FALSE	TRUE	TRUE	TRUE	FALSE	FALSE	Positive
P14	Swollen Lymph Nodes	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	Negative
P15	Swollen Lymph Nodes	FALSE	TRUE	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE	Positive
P16	None	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	Positive
P17	None	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	Positive
P18	Muscle Aches and Pain	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	Negative
P19	Swollen Lymph Nodes	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	Positive
P20	Fever	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	Negative
P21	None	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	Negative

Fig. 6 Snapshot form used dataset

Fig. 7 Heatmap displays the correlation between dataset features

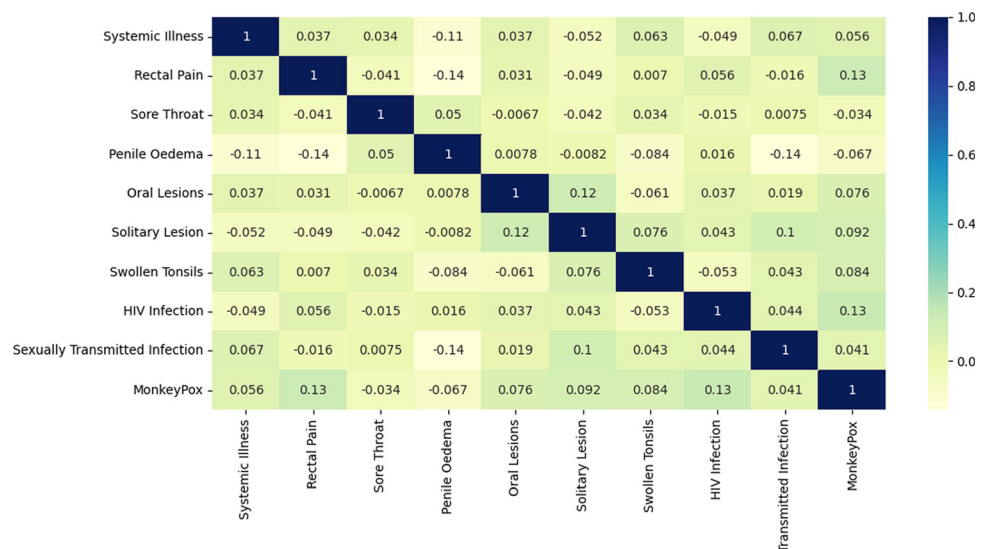


Table 1 Statistical methods for measuring the accuracy of a machine learning model. [42]

Method name	Equation
Accuracy	$\frac{T_p + F_p}{T_p + T_n + F_p + F_n}$
Precision	$\frac{T_p}{T_p + F_p}$
Sensitivity	$\frac{T_p}{T_p + F_n}$
F1-score	$2 * \frac{\text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$

Results from the experiments are shown in Fig. 5. ABC and aABC methods could not produce very successful solutions for the CEC01 function, but they reached the best

solution for the CEC02 function, like the other four algorithms. All compared algorithms for the CEC03 function achieved the best results. The best solutions for the CEC04 and CEC05 functions were obtained with aABC. The classical ABC method produced the third-best solution for the CEC04 function and the second-best solution after aABC for the CEC05 function. For functions CEC06 and CEC07, aABC is the second-best solution. In CEC09 and CEC10 functions, aABC produced the second-best solutions. In this case, aABC produced the best solution in five out of ten cases in the CEC2019 test suites. When the solution averages of the algorithms are examined, it is seen that aABC produces successful solutions across the functions. As a general evaluation, the success of the ABC algorithm in numerical optimization problems has been

Table 2 Performance of proposed model

Phase	Performance			
	Accuracy (%)	F1-score (%)	Precision (%)	Sensitivity (%)
Training	78	76	81	84
Testing	71	72	69	67

repeated, and the effect of aABC on the classical ABC method has been proven.

3.3 Dataset

The dataset based on a study published by the bmj: Clinical features and novel presentations of human monkeypox in a central London center during the 2022 outbreak: descriptive case series. [40] Features: Patient_ID, Systemic Illness, Rectal Pain, Sore Throat, Penile Edema, Oral Lesions, Solitary Lesion, Swollen Tonsils, HIV Infection, Sexually Transmitted Infection and Target Variable: MonkeyPox. The Monkeypox dataset comprises 240 cases classified into two categories: 'positive' and 'negative'.

The positive cases are monkeypox patients, the negative cases are those who do not have monkeypox. In fact, a negative case does not mean that he is in good health but does not have monkeypox. However, this data determines whether or not he only had monkeypox. Figure 6. Snapshot of the monkeypox dataset. This dataset consists of 11 features, which include the patient's clinical symptoms of inflammation, fever and others. These features are used to describe the symptoms that appear on the patient in order to indicate the condition of each patient.

We calculated the linear correlation coefficient between features (Fig. 7). We found that the data had varying correlations with each other. Also, the dataset does not contain

null values. We coded the words as follows: True: 1, False: 0. We also divided this data into two groups, 80% for training and 20% for testing. In general, the percentage of infected and non-infected cases was distributed in equal proportions between these two groups.

In fact, the monkeypox dataset was divided into 168 cases as a training set of data and 72 cases as a test set of data. The number of positive cases is 120 and the negative cases are 120. Figure 7 shows how the features relate to each other.

3.4 Performance evaluation

Statistical methods are used to measure the accuracy of classification algorithms. These methods contribute to determining the standardization of the applied algorithm such as: accuracy, precision, *F1*-score and sensitivity. In our dataset, Monkeypox can be classified as True Positive or True Negative if the individuals have been accurately classified. It can be classified as False Positive or False Negative if misdiagnosed. Specific statistical measures are detailed in Table 1.

3.4.1 Root mean square error (RMSE)

It is a standard method for measuring model error. It is also called loss function. Know its equation as follows:

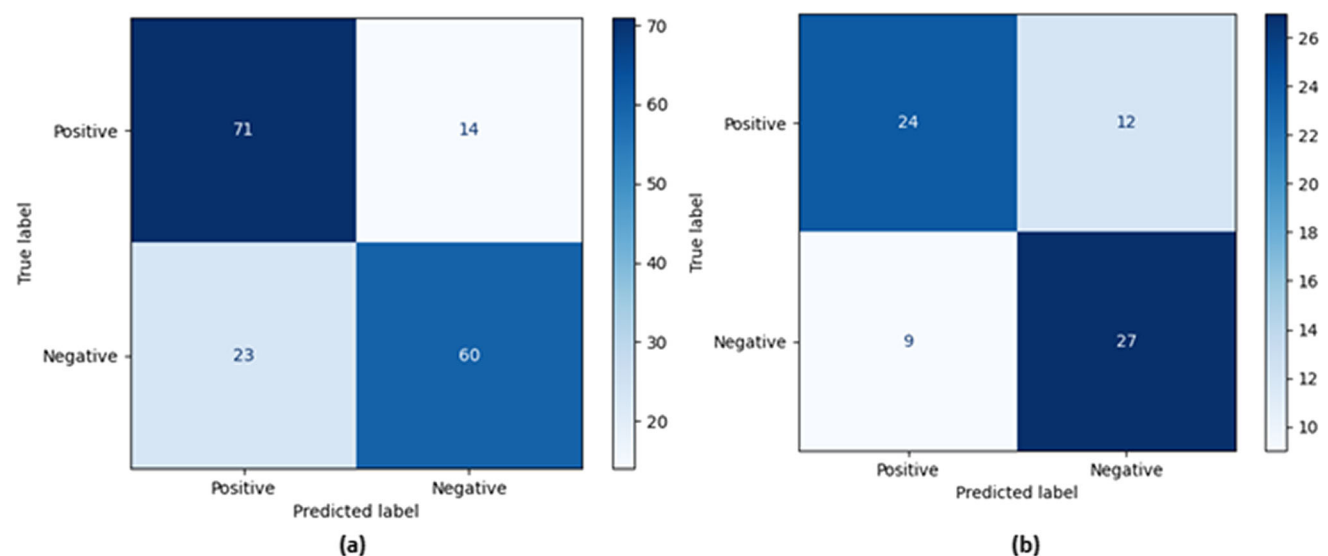


Fig. 8 **a** Confusion matrix of training phase for the proposed model. **b** Confusion matrix for testing

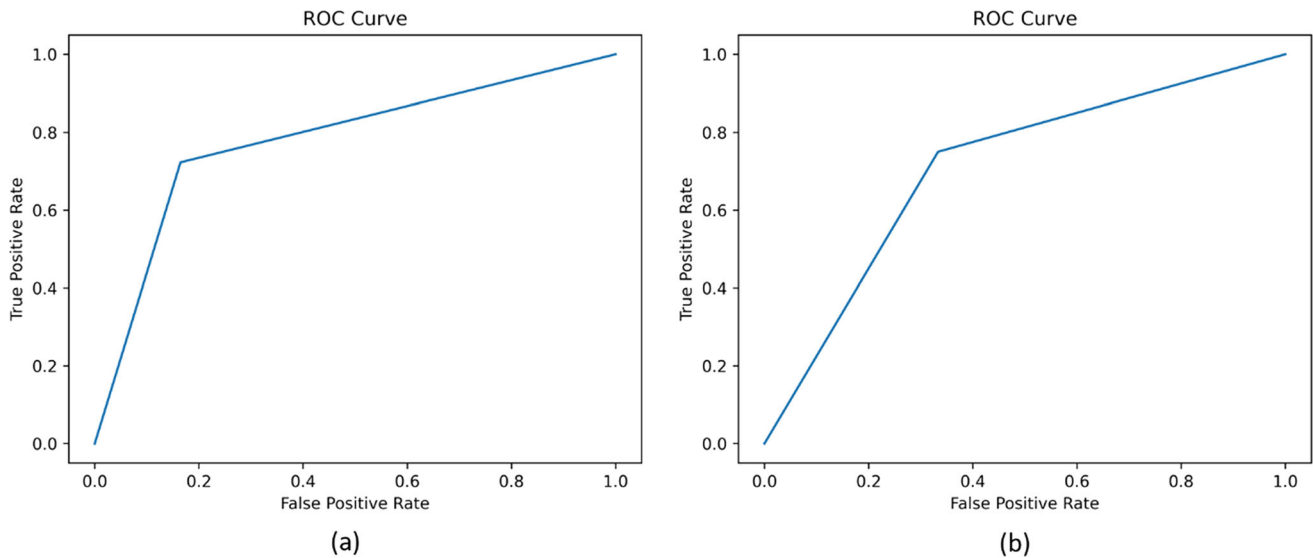


Fig. 9 **a** Represents AUC-ROC curve during training phase. **b** AUC-ROC during testing phase

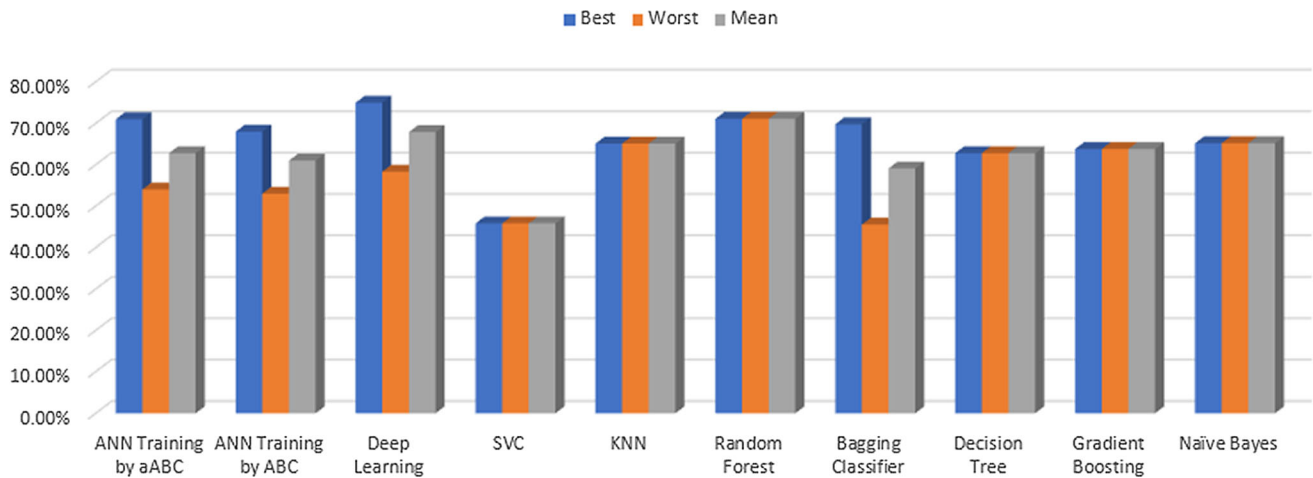


Fig. 10 Statistical results on the accuracy of all applied algorithms

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (y_o - y_p)^2}{N}} \quad (6)$$

The value of this equation tells us the distance difference between the vector of expected values and the vector of observed values. In data science, this formula is used to evaluate trained models. It gives us the error rate between the training results and original results.

3.4.2 Hyperparameter

The proposed model consists of four layers: one input layer, two hidden layers and one output layer. It contains 11, 10, 10, 1 neurons per layer, respectively. For the hidden layers, the activation function of the RELU Function has been set as in Eq. 7 and for the output layer, the activation

function of the Sigmoid Function has been set as in Eq. 8. As for the parameters of aABC algorithm, they are as follows:

Epoch = 400

food = 50

$$\text{Limit}[i] = \frac{\text{fitness}[i] * \text{food} * D}{\sum \text{fitness}[i]}$$

In order to avoid falling into the problem of (Vanishing gradient), the RELU function was used for the hidden layers. It is the basic way to solve this problem. However, to solve (Overfitting) K-Folds technique was relied in training the model.

$$y_j = f_j(x) = \max(0, x) \quad (7)$$

Table 3 Summarizes the performance of ten different models over 30 runs

Trials	Methods									
	ANN training by aABC	ANN training by ABC	Deep learning	SVC	KNN	Random forest	Bagging classifier	Decision tree	Gradient boosting	Naive Bayes
1	69.0%	64.0%	72.9%	45.8%	65.1%	71.1%	59.5%	62.8%	63.8%	65.2%
2	67.0%	62.0%	68.8%	45.8%	65.1%	71.1%	66.6%	62.8%	63.8%	65.2%
3	64.0%	62.0%	64.6%	45.8%	65.1%	71.1%	60.9%	62.8%	63.8%	65.2%
4	58.0%	61.0%	68.8%	45.8%	65.1%	71.1%	60.5%	62.8%	63.8%	65.2%
5	67.0%	57.0%	62.5%	45.8%	65.1%	71.1%	52.1%	62.8%	63.8%	65.2%
6	58.0%	58.0%	60.4%	45.8%	65.1%	71.1%	45.5%	62.8%	63.8%	65.2%
7	54.0%	67.0%	72.9%	45.8%	65.1%	71.1%	62.5%	62.8%	63.8%	65.2%
8	57.0%	56.0%	68.8%	45.8%	65.1%	71.1%	66.6%	62.8%	63.8%	65.2%
9	65.0%	57.0%	66.6%	45.8%	65.1%	71.1%	62.5%	62.8%	63.8%	65.2%
10	67.0%	65.0%	70.8%	45.8%	65.1%	71.1%	60.8%	62.8%	63.8%	65.2%
11	58.0%	61.0%	70.8%	45.8%	65.1%	71.1%	61.5%	62.8%	63.8%	65.2%
12	69.0%	65.0%	68.8%	45.8%	65.1%	71.1%	50.0%	62.8%	63.8%	65.2%
13	64.0%	65.0%	66.6%	45.8%	65.1%	71.1%	52.4%	62.8%	63.8%	65.2%
14	61.0%	60.0%	66.6%	45.8%	65.1%	71.1%	65.2%	62.8%	63.8%	65.2%
15	60.0%	53.0%	64.6%	45.8%	65.1%	71.1%	68.2%	62.8%	63.8%	65.2%
16	62.0%	67.0%	64.6%	45.8%	65.1%	71.1%	65.1%	62.8%	63.8%	65.2%
17	71.0%	57.0%	68.8%	45.8%	65.1%	71.1%	62.5%	62.8%	63.8%	65.2%
18	61.0%	60.0%	70.8%	45.8%	65.1%	71.1%	69.8%	62.8%	63.8%	65.2%
19	67.0%	61.0%	66.6%	45.8%	65.1%	71.1%	50.0%	62.8%	63.8%	65.2%
20	61.0%	54.0%	64.6%	45.8%	65.1%	71.1%	63.6%	62.8%	63.8%	65.2%
21	61.0%	65.0%	70.8%	45.8%	65.1%	71.1%	53.3%	62.8%	63.8%	65.2%
22	64.0%	64.0%	66.6%	45.8%	65.1%	71.1%	57.8%	62.8%	63.8%	65.2%
23	58.0%	62.0%	72.9%	45.8%	65.1%	71.1%	51.2%	62.8%	63.8%	65.2%
24	67.0%	57.0%	68.8%	45.8%	65.1%	71.1%	60.9%	62.8%	63.8%	65.2%
25	62.0%	65.0%	68.8%	45.8%	65.1%	71.1%	57.8%	62.8%	63.8%	65.2%
26	64.0%	56.0%	75.0%	45.8%	65.1%	71.1%	55.3%	62.8%	63.8%	65.2%
27	61.0%	64.0%	68.8%	45.8%	65.1%	71.1%	56.6%	62.8%	63.8%	65.2%
28	64.0%	58.0%	58.3%	45.8%	65.1%	71.1%	58.5%	62.8%	63.8%	65.2%
29	60.0%	68.0%	66.6%	45.8%	65.1%	71.1%	57.8%	62.8%	63.8%	65.2%
30	64.0%	58.0%	70.8%	45.8%	65.1%	71.1%	58.8%	62.8%	63.8%	65.2%
Best	71.0%	68.0%	75.0%	45.8%	65.1%	71.1%	69.8%	62.8%	63.8%	65.2%
Worst	54.0%	53.0%	58.3%	45.8%	65.1%	71.1%	45.5%	62.8%	63.8%	65.2%
Mean	62.8%	61.0%	67.9%	45.8%	65.1%	71.1%	59.1%	62.8%	63.8%	65.2%
SD	0.040	0.040	0.037	0.000	0.000	0.000	0.058	0.000	0.000	0.000

$$y_j = f_j(x) = \frac{1}{1 + e^{-x}} \quad (8)$$

4 Result and discussion

The proposal model is tested with a sample dataset consisting of 72 values taken from the dataset, it is a mixture of monkeypox cases collected at the bmj.

The sample data consisted only of cases of monkeypox with different symptoms. This also discusses the number of

positive cases and negative cases with different symptoms, thus proposing ANN model diagnostic method based on aABC algorithm. aABC is one of the evolutionary algorithms that contribute to the training of neural network weights.

Table 2 presents the results of the proposed model during training and testing. The performance of the model was measured by several criteria. They appear as follows: in the training period accuracy, *F1*-score, precision and sensitivity take the values 78%, 76%, 81%, 84%, respectively. When testing, accuracy, *F1*-score, precision and

sensitivity standards were taken as 71%, 72%, 69%, 67%, respectively. Figure 8 presents the confusion matrix of the proposed model during training and testing.

The performance of ten models of machine learning and deep learning algorithms and the proposed model are summarized in Table 2. All ten models were trained on the same dataset. The training and validation process for all algorithms was repeated 30 times, and the accuracy of each stage was recorded separately.

We also used another method to measure model performance called the AUC-ROC curve, as shown in Fig. 9. It is one of the evaluation tools approved in the classification. This graph measures the performance of the classification model when there are only two classes, a positive class and a negative class. This curve indicates a ROC curve that plots the False Positive Rate (FPR) on the horizontal axis (X) against the True Positive Rate (TPR) on the vertical axis (Y).

AUC-ROC is the area under this graph. If the area under the curve is large, this indicates good performance of the model in distinguishing between positive and negative classes. Therefore, we notice that the threshold value is close to 1, and therefore, the area covered by this curve is larger. Therefore, the model has good performance. This curve is found based on the confusion matrix values mentioned previously. Emphasis is also placed on positive and negative values in the model's performance.

Figure 9a represents a curve for the results of the training period, while figure (b) represents the results for the validation period. When the model's performance improves, the vertical axis values become close to 1, while the horizontal axis values become close to 0. For each of the two figures listed above, A and B.

Figure 10 confirms the superior performance of the RF algorithm compared to all other algorithms, with an average accuracy of 71.1%. (ANN) Deep learning model results secured the second position, achieving an average accuracy of 67.9%. Meanwhile, NB and KNN demonstrated commendable average performance. In terms of the highest accuracy, the (ANN) deep learning model took the lead with 75%, followed by the RF algorithm at 71.1%. While the model proposed in this paper showed the third-best performance, achieving an accuracy of 71%. However, the remaining seven models exhibited lower and unsatisfactory performance compared to (SVC), as indicated in Table 3. For other statistical metrics such as F1 score, sensitivity and accuracy, their ranges did not differ significantly from the accuracy measure.

The performance of our custom architectural model trained from scratch using aABC algorithm can be compared to other machine learning models. In fact, the proposed model could not outperform DL or even RF. While it was able to show better results than the rest. It is also noted

from Table 3 that the proposed model with DL, bagging classifier, changes the accuracy of each model every time it is tested, while the models such as SVC, KNN, RF, NB and decision tree maintained one result throughout the testing period. That is, algorithms that rely on a statistical principle are more stable than learning algorithms, although they did not outperform learning algorithms in their best performance, which is confirmed by the statistical measures in Table 3.

In this paper, they relied on collected dataset from people with the disease and suspects. They were collected at the bmj center. Our study differs from previous studies in this respect. This appears clearly if we look at the difference between the results obtained in this study and the results of previous research. Our goal was to obtain an early detection model of the disease using the clinical symptoms that appear on the person when suspected.

5 Conclusion

The paper provides a brief summary of the emergence of the monkeypox virus, a zoonotic disease transmitted from animals to humans. This virus belongs to the highly virulent Orthopoxvirus family. The spread of this disease in societies alarms many people. Therefore, society needs an automated system for early detection that helps detect infection with this disease if it occurs. Early prediction can prevent complications for people with the disease and save human lives. This study aims to provide a model for distinguishing monkeypox infection by the clinical symptoms associated with the disease that appear on the infected person. The proposed model hybridized aABC algorithm with ANN. Several models trained on the same training datasets were compared. The ANN deep learning model achieved the best performance with an accuracy of 75%, while the proposed model obtained an accuracy of 71%.

The proposed model is supported by several published studies that use an AI-based diagnostic model. We hope this article will contribute to future researchers and practitioners benefiting from the presented approach to develop a diagnostic mechanism for monkeypox disease. In the next study, we plan to prepare an AI method that can extract features for monkeypox using real-time data and classify them at a higher accuracy rate.

Appendix

See Table 4 .

Table 4 CEC2019 Benchmark functions [34]

No	Name	Area	Dimension
CEC01	Storn's Chebyshev polynomial fitting problem	[− 8192, 8192]	9
CEC02	Inverse Hilbert matrix problem	[− 16384, 16384]	16
CEC03	Lennard–Jones minimum energy cluster	[− 4, 4]	18
CEC04	Rastrigin's function	[− 10, 10]	10
CEC05	Griewangk's function	[− 10, 10]	10
CEC06	Weierstrass function	[− 10, 10]	10
CEC07	Modified Schwefel's function	[− 10, 10]	10
CEC08	Expanded Schaffer's F6 function	[− 10, 10]	10
CEC09	Happy cat function	[− 10, 10]	10
CEC10	Ackley function	[− 10, 10]	10

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Data availability The collected data is available as open data via an online data repository: <https://doi.org/https://doi.org/10.34740/KAGGLE/DSV/4271503>.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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